An Exact Bellman–Ford Transformer as Optimization Ground Truth

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Abstract

Hand-constructed transformers provide unusually strong ground truth: their forward pass, intermediate state, and intended circuit are known exactly. This does not imply that gradient descent can recover the same computation, even when the constructed solution belongs to the trainable model's hypothesis class. We study this prerequisite using a weight-tied transformer whose forward pass executes Bellman–Ford relaxation. The construction is not our novelty; its useful properties are exact verification, complete per-round state, and exact embedding into a trainable twin. On directed graphs with at most four nodes, a widened, LayerNorm-free, weight-tied model trained from random initialization reaches 96.97% held-out graph exactness, 98.85% node exactness, and 96.97% exactness on reachable non-source nodes. The embedded construction attains 100% exactness, zero objective, and zero gradient, yet the learned endpoint remains approximately as far from the constructed parameters as initialization. A matched short run is still improving at 50.20% graph exactness, while an 80,000-step run saturates near 97%. Perturbation experiments support optimizer-step drift rather than an objective that rewards loss of exactness. These are single-seed, small-graph diagnostics. We do not run the tied-versus-untied experiment needed to determine whether recurrence causes learned relaxation.

Keywords: algorithmic transformers, Bellman–Ford, weight tying, optimization, mechanistic interpretability

1 Introduction

Neural algorithmic reasoning asks whether a model has fit an input–output relation or learned a procedure that remains valid outside the training distribution. Shortest paths make the distinction concrete. A model can implement repeated local relaxation, exploit a correlated graph statistic, memorize a bounded-size mapping, or compute a different but behaviorally adequate algorithm. In-distribution output accuracy does not identify which explanation is correct.

Prior work establishes both sides of this problem. Shallow transformers can learn shortcut implementations of formal languages that fail under distribution shift [1]. A trained shortest-path transformer has been reported to use spectral navigation rather than textbook dynamic programming [2]. Conversely, recurrent transformers can learn iterative procedures with length generalization [3], and sparsity-regularized graph neural networks can recover Bellman–Ford-like relaxation [4]. The broad claim that trained models always fail to learn algorithms is therefore both occupied and false.

A controlled comparison needs a prerequisite stronger than ordinary task accuracy: the intended solution must be present in the trainable hypothesis class. Otherwise, failure to learn the intended circuit is uninterpretable. We call this requirement **G1 parity**. A trainable model has G1 parity when the hand-constructed parameters load into it and reproduce exact outputs and zero objective. This test rejects apparently minor architectural changes—most notably LayerNorm here—that alter the literal arithmetic performed by the construction.

This paper reports the diagnostic work required to establish G1 parity and learnability in one model family. Its contributions are deliberately bounded:

1. We present an exact Bellman–Ford transformer as executable optimization ground truth. The construction follows existing expressivity work; the contribution is the bit-exact verifier, complete round state, and direct embedding into a trainable twin.
2. For one seed and graphs with at most four nodes, a G1-compatible model learns near-exact behavior from random initialization. This separates representability from the optimization failure observed in shorter runs.
3. We characterize local drift around the exact circuit. The construction is a zero-loss, zero-gradient global minimum, and learning-rate-controlled runs recover exact behavior without recovering the original parameters.

The claim boundary is central. We do *not* show that weight tying makes a trained transformer use Bellman–Ford relaxation. We did not run an untied comparator, size extrapolation, state probes, attention comparison, iteration ablation, or causal intervention. Those are requirements for a later mechanistic experiment, not results of this study.

2 Related Work

2.1 Compiled and looped transformers

RASP provides a programming abstraction for transformer computation [5]. Tracr compiles restricted programs into transformer weights, producing models with known internal structure [6]. Looped-transformer constructions show that a reused block can implement iterative programs [7]. Most directly, Back de Luca and Fountoulakis [8] construct looped transformers for graph algorithms and discuss finite-precision and softmax limitations. These works preclude a construction-novelty claim for our model.

Our distinction is engineering and measurement discipline. The artifact exposes all residual registers after every relaxation, supplies an executable Bellman–Ford reference trajectory, and is tested for exact agreement rather than approximate task accuracy. The same parameter tensors embed into a widened trainable module with all added coordinates set to zero.

The implementation uses two different exactness mechanisms. Pairwise minima use the identity

$$\min(a,b) = a - \mathrm{ReLU}(a-b). \tag{1}$$

Routing uses a fixed float32 score gap large enough that non-target exponentials underflow to zero. The latter is an implementation property dependent on dtype and kernel behavior, not exact hard selection by finite-logit real-arithmetic softmax. Approximate sharpness of ordinary softmax requires score spread growing with set size and desired error [9].

2.2 Learned algorithms and shortcuts

Liu et al. [1] show that low-depth transformers can fit automata through shortcuts that fail outside the training regime. On graphs, Cohen et al. [2] analyze a trained shortest-path transformer and report a spectral mechanism rather than Bellman–Ford or Dijkstra-style dynamic programming. These results own the broad shortcut-learning headline.

The negative conclusion is not universal. Looped transformers have been trained to learn iterative solutions with length generalization [3]. Sparsity-regularized GNNs can recover a fixed number of Bellman–Ford iterations and extrapolate to larger graphs [4]. MINAR recovers compact Bellman–Ford-like circuitry in a trained GNN and also demonstrates that correct outputs can conceal an unintended internal shortcut [10].

The remaining narrow question is whether weight-tied recurrence changes what a trained transformer learns when an exact relaxation circuit exists in the same architecture. The present study validates a prerequisite for asking that question, but does not answer it.

2.3 Known-circuit interpretability

InterpBench argues that mechanistic-interpretability methods need models with known circuits beyond raw Tracr programs [11]. A hand-constructed graph algorithm offers exact labels at the level of registers and iterations, but one model is not itself a benchmark. Establishing benchmark value would require comparing interpretability methods or exposing a specific failure against the known circuit.

Intermediate-state probes would also be insufficient alone. Candidate algorithmic state can be decodable without being causally used [12]. A future claim that a trained model executes relaxation must therefore combine state alignment with interventions.

3 Exact Transformer Artifact

3.1 Algorithm and state

For source s, Bellman–Ford initializes

$$d_s^{(0)} = 0, \qquad d_v^{(0)} = \mathrm{INF}_B \quad (v \neq s), \tag{2}$$

where $\mathrm{INF}_B = N_{\max} W_{\max} + 1$ exceeds every representable simple-path length. Each round computes

$$d_v^{(t+1)} = \min\left[d_v^{(t)}, \min_{(u,v,w)\in E} \left(d_u^{(t)} + w \right) \right]. \tag{3}$$

The sequence contains one sink token, a fixed allocation of node tokens, and a fixed allocation of directed-edge tokens. Residual coordinates encode node identities, the distance register, edge endpoints and weights, candidate slots, and temporary routing state. A single block performs one relaxation (Figure 1): edge tokens read source-node distance; gather heads route incoming candidates to destination slots; a ReLU network folds those candidates with the previous distance; and a residual delta writes the result while clearing scratch state. The same block is reused $N_{\max} - 1$ times. An optional extra round detects a reachable negative cycle by checking whether a distance decreases.

3.2 Verification and tracing

The verifier independently computes every Bellman–Ford reference state. Integer-weight outputs are bit-exact against NetworkX's single-source Bellman–Ford implementation; a seeded 1,000-graph property suite spans graph sizes through the configured maximum. The tracer records the full residual stream, named sublayer deltas, and attention patterns.

This verification does not make the construction novel. It makes the construction an exact reference point for optimization and later mechanistic comparison.

4 Trainable Twin and G1 Parity

The trainable twin preserves tokenization, residual-register layout, sublayer order, head topology, and recurrent application. Attention projections and MLP hidden spaces may be widened. To load the construction, each original matrix is copied into the leading block of the wider tensor and every additional entry is set to zero. Widening therefore preserves the exact constructed function.

LayerNorm is optional in diagnostic variants but incompatible with G1 parity: normalizing the residual stream rescales literal distances. A G1 audit requires a LayerNorm-free model, an exact constructed-weight load, exact decoded outputs, and zero objective.

The widened LayerNorm-free model passes this audit. Validation and test graph, node, reachability, and reachable-non-source exactness are all 100%; supervised loss and MAE are zero; the full-objective gradient has zero ℓ_2 norm and zero maximum absolute component; and Euclidean distance from the constructed reference is zero. Since the objective is non-negative, zero objective certifies that the construction is a global minimum of the measured loss.

5 Diagnostic Protocol

All results are diagnostic. Completed runs use model seed 17 and data seed 20260730; the fixed validation and test sets contain 512 and 1,024 graphs. Training graphs are freshly sampled on every batch. Graphs are directed, have integer non-negative edge weights, and contain two to four nodes. The model executes three tied rounds.

The principal random-initialization recipe uses width 128 for every attention and MLP subspace, no LayerNorm, per-round supervision, batch size 128, and Xavier initialization with gain 0.1. Optimization uses AdamW with $\beta = (0.9, 0.999)$, zero weight decay, and gradient norm clipped to 1. The learning rate warms up for 200 steps to 3×10^{-4}, then decays by a cosine schedule to zero over 80,000 steps.

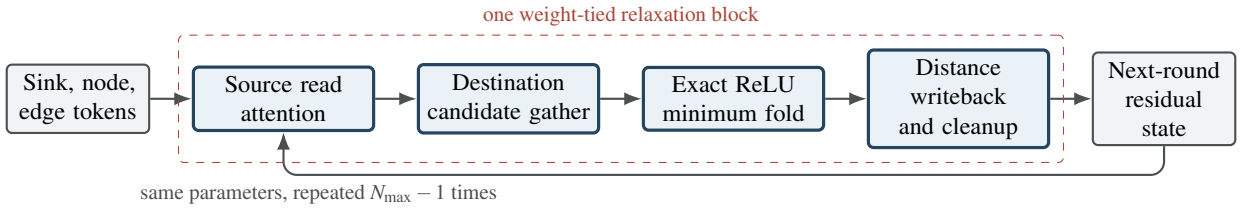


Figure 1. The constructed residual program. Attention performs fixed routing; the ReLU fold performs exact arithmetic minimum; the full block is reused for each Bellman–Ford round.

5.1 Objective and exactness

The objective is mean Smooth-L1 loss over valid nodes and supervised rounds:

$$\mathscr{L} = \text{mean}\,\text{SmoothL1}_{\beta=0.01}\left(\hat{d}/\text{INF}_B,\ d/\text{INF}_B\right). \qquad (4)$$

For finite targets, a node is exact if $|\hat{d} - d| < 0.5$. For unreachable targets, it is exact if $\hat{d} \geq \text{INF}_B - 0.5$. Graph exactness requires every real node to be exact. We additionally report node exactness, reachability accuracy, and reachable non-source exactness (RNSE). RNSE removes source and unreachable nodes, the cases already correct under identity-like initialization.

5.2 Saturation criterion

No inherited accuracy threshold defines competence. A run is saturated only if its terminal learning rate is zero and graph exactness, node exactness, and RNSE each satisfy three conditions over the final two 10,000-step windows: (i) final-window best improves by less than one percentage point over the preceding window; (ii) absolute least-squares slope is below one point per 10,000 steps; and (iii) final-window range is below two points. This establishes a measured plateau, not global optimality.

5.3 Local perturbation diagnostic

To separate loss–metric conflict from optimizer drift, we start from the exact constructed parameters plus independent Gaussian noise with standard deviation 4×10^{-4}. These runs use the constructed-minimal, tied, LayerNorm-free model. We measure objective and gradient at the construction, then log validation loss, exactness, and Euclidean parameter distance. Peak learning rates 3×10^{-5} and 10^{-5} both decay to zero.

6 Results

6.1 G1-compatible training succeeds at $n \leq 4$

Table 1 reports the saturated random-initialization runs. The LayerNorm-free model reaches 96.97% held-out graph exactness and 96.97% RNSE. This establishes a trainable, G1-compatible recipe in the tested regime. LayerNorm is not necessary under the widened, long-horizon recipe. The 0.78-point validation graph difference is a single-seed observation, not an estimate of a systematic normalization effect.

The trained no-LayerNorm endpoint is functionally successful but does not approach the constructed vector. Its Euclidean distance from the construction is approximately 76,648, versus approximately 76,649 at initialization. The evidence establishes functional learnability, not literal recovery of the known parameterization.

6.2 The matched short run is budget-limited

Table 2 compares otherwise matched LayerNorm runs. The 4,000-step condition ends at its best checkpoint and is not a ceiling. Extending the schedule to 80,000 steps changes validation graph exactness by 47.27 points.

This intervention could have falsified a budget effect under the selected recipe and instead found a large one. It does not show that duration was the sole cause of earlier failure. The 4,000-step diagnostic already exceeds the earlier approximately 18% post-initialization graph exactness, so another difference also mattered. Fresh versus fixed data, width 128 versus constructed-minimal width, per-round versus final-only supervision, and batch 128 versus batch 32 remain untested as isolated causes.

6.3 Drift is consistent with optimizer step size

At the exact construction, the objective and both reported gradient norms are zero. The loss therefore does not locally push the exact circuit away. After perturbation, both zero-floor schedules recover exact or nearly exact behavior (Table 3). The smaller peak learning rate ends at 100% validation and test graph exactness, although parameter distance grows slightly instead of returning to zero.

Across both logged trajectories, there are zero evaluation intervals in which validation graph exactness falls while validation loss also falls. Exactness declines coincide with higher validation loss, not an objective improvement. Both endpoints have positive loss; the construction has zero.

These measurements support an ordinary optimizer-step explanation for the previous drift: updates can leave a narrow exact region, and reducing peak learning rate limits that movement. They do not support the alternative claim that the regression objective rewards sacrificing half-unit-bin exactness. Parameters need not return to the construction to recover the measured function.

7 Discussion

7.1 Representability is executable

It is easy to describe a model as having "failed to learn the known algorithm" when the intended circuit cannot be represented by the training architecture. G1 parity turns that ambiguity into a test: load the construction, verify exact outputs, and verify zero objective. LayerNorm fails because it changes the arithmetic state. Widening passes when the original matrices are embedded and all new coordinates are zeroed. This is a practical compromise: optimization is not restricted to constructed-minimal width, but the exact circuit remains in the hypothesis class.

Table 1. Saturated random-initialization runs. Exactness values are percentages. The test reachability accuracy is 100% for both runs.

Architecture	G1	Best step	Validation graph / node / RNSE	Test graph / node / RNSE	Verdict
Width 128, pre-LayerNorm	no	70k	**97.46 / 99.14 / 97.64**	96.97 / **98.92** / 97.14	saturated
Width 128, no LayerNorm	yes	80k	96.68 / 98.74 / 96.56	**96.97** / 98.85 / **96.97**	saturated

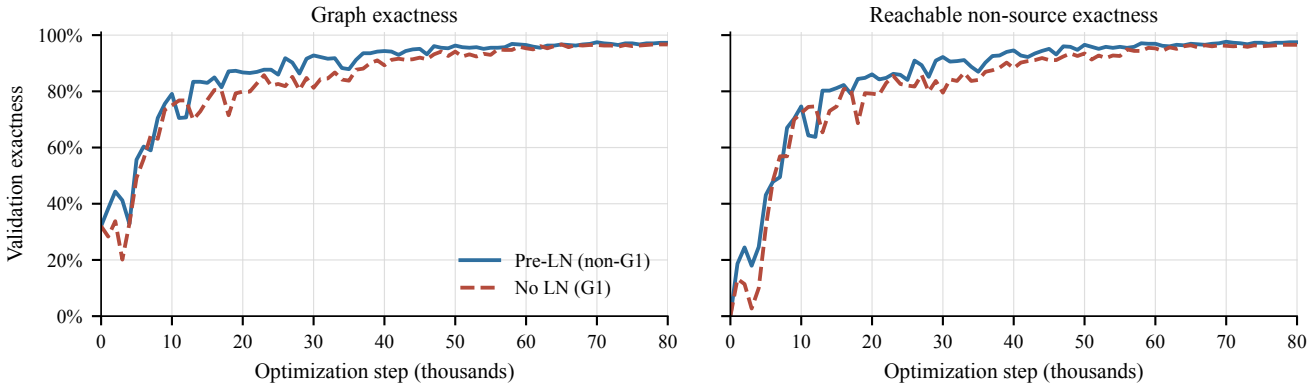


Figure 2. Validation histories, logged every 1,000 optimization steps, from the frozen v4 artifact. Both width-128 models continue improving well past the original 4,000-step budget. The no-LayerNorm run preserves G1 parity.

Table 2. Matched training-horizon intervention.

Steps	Graph	Node	RNSE	Curve
4k	50.20	65.25	27.72	improving
80k	97.46	99.14	97.64	saturated

Table 3. Constructed-basin perturbation diagnostics.

Peak LR	Val. graph	Test graph	Distance $i \to f$
3×10^{-5}	100.00	99.80	$0.0355 \to 0.1257$
1×10^{-5}	100.00	100.00	$0.0355 \to 0.0433$

Final losses are 7.06×10^{-7} and 7.58×10^{-8}, respectively; the construction has loss zero.

7.2 Functional success is not circuit recovery

The random model learns a near-exact function while remaining almost as far from the constructed vector as at initialization. Local perturbation runs recover exact outputs without returning to the reference parameters. Raw parameter distance is therefore not a valid circuit-correspondence criterion here.

This observation does not establish that the learned mechanism differs from Bellman–Ford. Neural parameterizations have symmetries and redundant representations, while widening adds alternative subspaces. Mechanistic correspondence must be tested at the level of round state, routing, intervention response, and executed function.

7.3 What is known about failed short runs

The supported causal claim is narrow: under the selected recipe, 4,000 steps are insufficient and 80,000 steps reach saturation. The evidence does not show that data reuse, minimal capacity, missing LayerNorm, supervision, or batch size caused a prior failure. Those explanations remain hypotheses because the corresponding one-factor interventions were not completed.

The distinction is methodological. A checkpoint at the end of an improving curve supports "budget-limited," not "unlearnable." Conversely, a high score under a favorable recipe does not identify which ingredient repaired an earlier run.

8 Limitations and Claim Boundary

The limitations determine the contribution:

- All completed training results use one model seed. There are no seed variances, confidence intervals, or population-level comparisons.
- Training and evaluation stop at $n \leq 4$. An $n \leq 8$ saturation run was not completed, and no graph-size extrapolation was evaluated.
- The principal recipe uses per-round supervision, fresh graphs, width 128, batch 128, and 80,000 steps. Most ingredients were not isolated.
- The tied-versus-untied and final-only-versus-per-round grid was not run. Nothing here determines whether recurrence changes the learned mechanism.
- No probe matrix, shuffled-label baseline, attention agreement, iteration ablation, or causal state intervention was run.
- Parameter distance depends on parameterization, symmetries, and widening. It rejects literal vector recovery, not functional equivalence.
- Exact attention depends on tested float32 underflow behavior and is not a real-arithmetic softmax theorem.
- Saturation describes measured final windows; it does not prove that no better optimizer or longer schedule could improve the result.
- These diagnostics followed earlier failed pilots and are not a preregistered confirmatory experiment.

This is workshop-scale evidence for experimental validity and optimization behavior. It is not a shortest-path generalization result and not a mechanistic claim about learned relaxation.

9 Deferred Mechanistic Experiment

The exact artifact makes a stronger study possible, but that study remains future work. A valid test must first reproduce G1-compatible learnability at the intended $N_{\max} = 16$ scale, train on $n \leq 8$, and evaluate through $n = 16$. It would then compare tied and untied models under final-only and per-round supervision

across multiple seeds.

The decisive measurements are: (i) behavioral exactness stratified by size and shortest-path hop count; (ii) a full $k \times j$ matrix probing loop state at iteration k for Bellman–Ford state after round j, with random and shuffled-label baselines; (iii) attention-routing agreement with the construction; (iv) iteration ablation; and (v) causal interventions on decoded round state, performed only if a probe diagonal appears.

Without these conditions, the question "does weight-tied recurrence make a trained model learn relaxation?" remains unanswered.

10 Conclusion

An exact algorithmic transformer supplies more than an expressivity example: it gives an executable global minimum, exact intermediate state, and a direct test of whether a trainable architecture contains the intended solution.

In the small-graph regime, a widened, LayerNorm-free, weight-tied model that contains the exact Bellman–Ford circuit reaches 96.97% held-out graph exactness from random initialization. The trained parameters do not recover the construction. Local perturbation experiments show that exact functional recovery depends on optimizer step size while remaining compatible with parameter drift.

This resolves an experimental-validity objection, not the motivating mechanistic question. Whether recurrence changes a trained transformer's algorithm remains open because the required comparison was not run.

Reproducibility

Complete evaluation histories are frozen in `docs/pilot-v4-curves.json`; the audit trail is in `docs/pilot-v4-results.md`. Diagnostic implementation is in `train/v3_model.py`, `train/v3_diagnostic.py`, and `train/v4_diagnostic.py`. Source commit `6489e31` contains the v4 diagnostics; `ccb199b` freezes the report and curves. Runs used Python 3.11.9, PyTorch 2.11.0+cu126, and an NVIDIA RTX 3090.

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A Evidence Ledger

Table 4 separates supported conclusions from unresolved causes. This ledger is part of the claim boundary, not a list of proposed explanations.

B Additional Completed Runs

The primary text reports the two saturated width-128 endpoints and the matched budget intervention. Table 5 records the remaining completed random-initialization search. The 60k width-128, 3×10^{-4} run is explicitly budget-limited because graph and RNSE late-window gains exceed the saturation bound. Width 256 saturates below width 128 for this seed, falsifying the simple claim that additional width monotonically repairs optimization. The 10^{-3} run is also budget-limited at 60k and therefore does not establish a lower asymptotic ceiling.

Table 4. Evidence ledger. "Untested" means no causal conclusion is drawn.

Question	Status	Evidence
Does the trainable no-LN model contain the exact circuit?	Established	Constructed embedding gives 100% exactness, zero objective, zero gradient, and zero parameter distance.
Can that architecture train from random initialization at $n \leq 4$?	Established for one seed	Saturated held-out graph exactness is 96.97%; held-out RNSE is 96.97%.
Is LayerNorm necessary under the selected recipe?	No, for one seed	The no-LN run saturates within 0.78 validation graph points of pre-LN and ties its test graph exactness.
Does training duration matter under the selected recipe?	Established	The matched 4k run is still improving at 50.20%; the 80k run saturates at 97.46% validation graph exactness.
Why did the original short pilot fail?	Unresolved	Budget matters, but the matched 4k diagnostic already exceeds the earlier result. Fixed data, minimal width, final-only supervision, and batch size were not isolated.
Does descent prefer lower loss at worse exactness?	Not supported	The construction has zero loss and gradient. No logged interval shows validation loss and exactness falling together.
Does training recover the constructed parameters?	No literal recovery	The random endpoint remains approximately as far from the constructed vector as initialization; local exact recovery also ends at nonzero distance.
Does weight tying cause learned relaxation?	Unanswered	No untied comparator, size extrapolation, probe matrix, iteration ablation, or causal intervention was run.

Table 5. Additional random-initialization diagnostics. Values are best validation graph exactness / RNSE.

Width	LN	LR	Horizon	Graph / RNSE
128	yes	3×10^{-4}	60k	96.48 / 96.20
128	yes	3×10^{-4}	80k	97.46 / 97.64
256	yes	3×10^{-4}	60k	94.92 / 94.75
128	yes	1×10^{-3}	60k	88.09 / 86.41
128	no	3×10^{-4}	80k	96.68 / 96.56

C Exclusions and Operational Record

No $n \leq 8$ result is reported. A blocking remote run lost its SSH session at step 49k before writing a result artifact; a detached replacement was stopped under the fixed final time budget. Neither reached the end of its learning-rate schedule, so neither establishes saturation or a ceiling.

A first 10^{-3} attempt overlapped two other jobs on the same physical RTX 3090 and terminated out of memory; it was rerun alone. Two fleet aliases were found to resolve to that same host. A fixed-data run that overlapped another job before the alias was discovered was stopped and excluded. Interrupted runs are absent from the frozen curve artifact. These are operational exclusions, not additional observations.

D Build Recipe

The plot in Figure 2 is generated directly from the frozen JSON by `paper/plot_figures.py`. The manuscript builds with pdfLaTeX and BibTeX; all plots remain vector graphics. The repository test suite contains 47 tests, including the 1,000-graph exactness property suite, and passes at the paper commit.

E Meaning of "Exact"

This study keeps four notions separate. First, the *constructed algorithmic output* is bit-equal to the integer Bellman–Ford reference in the tested float32 environment. Second, *trained-model exactness* is the registered task metric: a finite prediction lies within one half unit of its integer target, while an unreachable prediction lies above $\mathrm{INF}_B - 0.5$. Thus, the reported 96.97% is graph-level metric exactness, not bit equality of every trained activation.

Third, *attention routing exactness* is caused by float32 exponential underflow at a fixed score gap. It is conditional on dtype and kernel behavior; finite real-arithmetic softmax remains strictly positive on every candidate. Fourth, **G1 exactness** means that zero-padding the constructed weights into the widened no-LN model preserves the constructed function, objective, and gradient exactly. None of these statements is a general softmax exactness theorem.